FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2021

TEST DATE: 23-12-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A telescope has an objective of focal length 100 cm and an eyepiece of focal length 2.5 cm. If you look into the objective (that is, into the wrong end) of this telescope, you will see distant objects reduced in size. By what factor will the angular size of objects be reduced?
   (A) 40
   (B) 10
   (C) 20
   (D) 4

2. Consider a body moving along a circular path of radius R. Given some statements:-
   (a) The acceleration of the body is directed towards the centre
   (b) The acceleration of the body is directed towards the tangent.
   (c) One of the forces acting on the body must be variable in nature.
   (d) The radial acceleration of the body is non-zero.
   Which of the statements are always true?
   (A) a,b,d
   (B) a,b,c,d
   (C) b,c,d
   (D) c,d

3. Assume that you are facing south direction at a time when the sun is just setting. The scattered light that comes to you is polarized in a plane which is
   (A) Horizontal with electric field along vertical direction.
   (B) Vertical with electric field along east west direction.
   (C) Vertical with electric field along vertical direction.
   (D) Horizontal with electric field along east west direction.

4. Light with wavelength 500 nm falls on a slit of width 10 μm at an angle 30° to its normal. Find the angular position of the first minima located on both sides of the central Fraunhofer maximum:-
   (A) 33° and 27°
   (B) 25° and 35°
   (C) 36° and 24°
   (D) 28° and 32

5. Let A_n be the area enclosed by the nth orbit in a hydrogen atom. The graph of $\ln (A_n / A_1)$ against $\ln (n)$
   (A) Will not pass through theorigin
   (B) Will have certain points lying on a straight line with slope 4
   (C) Will be a monotonically increasing non linear curve
   (D) Will be a circle

6. Diatomic gas is used in Carnot heat engine. If efficiency of given Carnot heat engine is 80%, the find the ratio of initial volume to final volume of gas during adiabatic expansion:

(A) $\left(\frac{1}{5}\right)^{3/2}$

(B) $\left(\frac{1}{3}\right)^{5/2}$

(C) $\left(\frac{1}{5}\right)^{5/2}$

(D) $\left(\frac{1}{5}\right)^{2/5}$

7. A spring mass system executes damped harmonic oscillations given by the equation

$y = Ae^{-\frac{bt}{2m}}\sin\left(\omega' t + \phi\right)$

Where the symbols have their usual meanings. If a 2kg mass (m) is attached to a spring of force constant (K) 1250 N/m, the period of the oscillation is $\left(\pi/12\right)s$. The damping constant 'b' has the value.

(A) $9.8\,\text{kg/s}$

(B) $2.8\,\text{kg/s}$

(C) $98\,\text{kg/s}$

(D) $28\,\text{kg/s}$

8. When a screw gauge is completely closed, zero of circular scale is 7 divisions above the reference line of graduation when the zero of main scale is not seen. If LC of screw gauge is 10^{-3} cm, the zero error is

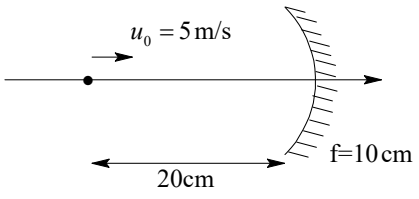
(A) -7×10^{-3} cm

(B) $+7 \times 10^{-3}$ cm

(C) –0.007 mm

(D) + 0.007 mm

9. An object moves with a uniform velocity u_0 along the axis of a concave spherical mirror. Consider the instant shown in diagram, object is moving with u_0 = 5 m/s and f = –10 cm. If object is at the centre of curvature at this instant then the magnitude of acceleration of image at this instant is a m/s^2.



Find a.

(A) 10

(B) 5

(C) 25

(D) None of these

10. The coil in a moving coil Galvanometer experiences torque proportional to current passed through it. If a steady current i is passed through it the deflection of the coil is found to be $90°$. Now the steady current is switched off an a charge Q is suddenly passed through the coil. If the coil has N turns of area A each and its moment of inertial is I about the axis it is going to rotate then the maximum angle through which it deflects upon passing Q is:

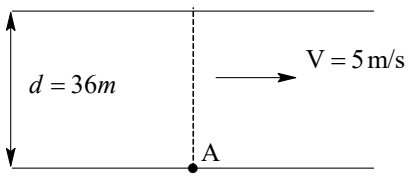
(A) $Q\sqrt{\frac{\pi NAB}{2iI}}$

(B) $\frac{\pi Q}{2}\sqrt{\frac{NAB}{iI}}$

(C) $\pi Q\sqrt{\frac{NAB}{2iI}}$

(D) $Q\sqrt{\frac{\pi NAB}{iI}}$

11. A boat starts from one bank of the river from point A to reach other by crossing width d of the river with minimum drift. Speed of a boat with respect to river is 4 m/s and speed of river with respect to ground is 5 m/s. Value of d = 36m, then



(A) Minimum drift will be zero.
(B) Minimum drift will be 27m.
(C) Time taken for minimum drift will be 20 sec.
(D) Angle at which boat will be steered from river flow direction for minimum drift is 127°.

12. A pipe of length 1.5 m closed at one end is filled with gas and resonates in its fundamental mode with a tuning fork. Another open organ pipe of same dimensions filled with air resonates in its fundamental mode with same tuning fork. If experiment is performed at $30°$ C (speed of sound in air Is 360 m/sec at $30°$ C), the speed of sound at 0°C in gas is (Neglect end correction):

(A) 580 m/sec.
(B) 683 m/sec.
(C) 880 m/sec.
(D) 743 m/sec.

13. A uniform string suspended vertically. A transverse pulse is created at the top most of the string. Then

(A) Speed of pulse remains constant.
(B) The speed of the pulse decreases with constant rate as pulse moves downward.
(C) The speed of the pulse decreases with increasing rate as pulse moves downward.
(D) The speed of the pulse increases with constant rate as pulse moves downward

14. An N-P-N transistor in a common emitter mode is used as a simple voltage amplifier with a collector current of 4mA. The terminal of a 8V battery is connected to the collector through a load resistance R_L and to the base through a resistance R_B. The collector emitter voltage V_{CE} = 4V, base-emitter voltage V_{BE} = 0.6 V and base current amplification factor β DC = 100. Calculate the value of R_L and R_B.

(A) $R_L = 1k\Omega$, $R_B = 185\ k\Omega$
(B) $R_L = 2k\Omega$, $R_B = 150\ k\Omega$
(C) $R_L = 1k\Omega$, $R_B = 240\ k\Omega$
(D) $R_L = 3k\Omega$, $R_B = 185\ k\Omega$

15. In a single slit diffraction pattern, if first maxima of light of wavelength λ coincides with first minima of yellow light (540 nm), then λ is:
(A) 270 nm
(B) 360 nm
(C) 540 nm
(D) 810 nm

16. If there is an error of 1% in calculation of mass of disc & 1.5 % error in radius, then % error in moment of inertia about an axis tangent to disc is:
(A) 2.5%
(B) 4%
(C) 3.5%
(D) 5%

17. To convert galvanometer into ammeter, shunt of $0.01\,\Omega$ is used. Resistance of galvanometer coil is $50\,\Omega$ and its maximum deflection current is 20 mA. Range of ammeter is:
(A) 0-1 A
(B) 0-10 A
(C) 0-100 A
(D) 0-1000 A

18. Unpolarized light falls on two polarizing sheets placed one after another. What should be the angle between pass axis of the sheets so that final transmitted light have one fourth intensity of incident light?
(A) 30°
(B) 0°
(C) 60°
(D) None of these

19. A hydrogen atom at rest emits a photon during its transition from n = 2 to n = 1. Choose the **INCORRECT** statement.
(A) De-Broglie wavelength of hydrogen atom during recoil will be same as wavelength of emitted photon.
(B) Energy of emitted photon will be less than 10.2eV
(C) Kinetic energy of recoil H-atom will be less than 10.2eV
(D) Kinetic energy of recoil H-atom will be equal to energy of emitted photon

20. A wheel of radius r starts rolling with an angular velocity ω and initial linear velocity V_0 up and inclined smooth plane. The wheel will stop going up in time:
(A) $\dfrac{V_0}{g\sin\theta}$
(B) $\dfrac{r\omega}{g\sin\theta}$
(C) $\dfrac{V_0 - r\omega}{g\sin\theta}$
(D) $\dfrac{r\omega - V_0}{g\sin\theta}$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A particle executes SHM along a straight line with mean position at x = 0, time period 12 second and amplitude 5cm. Find the minimum time (in second) in which particle will move from $x = -3\text{cm}$ to $x = 4\text{cm}$

22. Concrete sphere of radius R has a cavity of radius *r* which is packed with sawdust. The specific gravities of concrete and sawdust are respectively 2.4 and 0.3. For this sphere to float with its entire volume submerged under water, ratio of mass of concrete to mass of sawdust will be:-

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A paramagnetic sample shows a net magnetization of $8\,Am^{-1}$ when placed in an external magnetic field of $0.6T$ at a temperature of 4K. When the same sample is placed in an external magnetic field of 0.2T at a temperature of 16K, the magnetization $\left(\text{in Am}^{-1}\right)$ is:

24. A particle moving with constant speed turns left by an angle of $74°$ after travelling every 1 m distance. If it returns back to its starting point after 18 sec, then the magnitude of average acceleration of the particle from its initial position to just before it takes its second turn is 50 k m/s^2. Find the value of K.

25. The electrostatic flux through a rectangular area of $\frac{7}{188}\text{m}^2$ lying in a plane 2x + 3y + 6z = 10 due to an electric field $\left(8\hat{i}+6\hat{j}+10\hat{k}\right)\text{V/m}$ will be in (V-m)

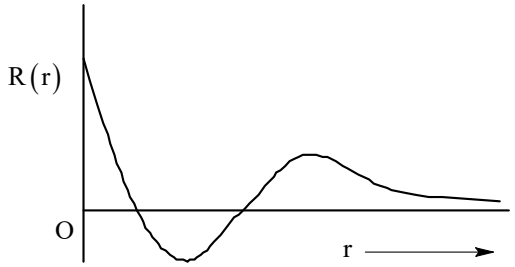
Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. The radial part of wave function of an orbital is plotted against distance from nucleus. Which orbital represent below graph?

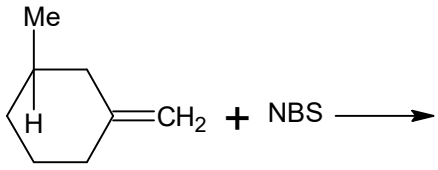


(A) 1s
(B) 2s
(C) 3s
(D) 2p

27. Dilute solution of KOH reacts with I_2 and forms:

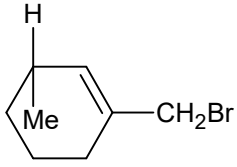
(I) KI (II) KI_3
(III) KIO

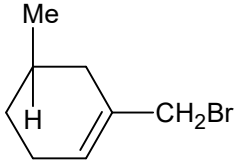
(A) II,II
(B) I,III
(C) I
(D) I,II,III

28. 

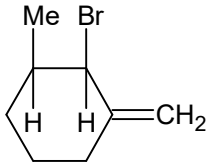
six procucts:

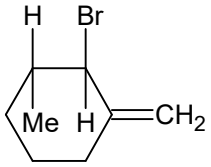
Which of the following statements is incorrect about products?

(A) 

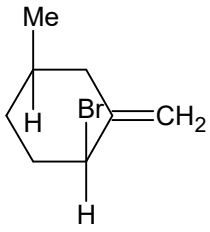
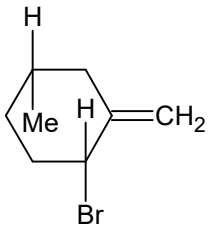
, 

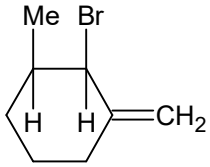
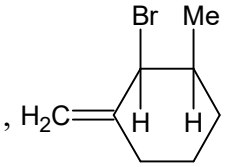
are two possible products.

(B) 

, 

are other two possible products

(C)  ,  are other two possible products.

(D)  ,  are other two possible products

29. $0.42g$ of Mg_2C_3 reacts with water produces gas X. The gas X combination with O_2 produce CO_2. The volume of CO_2 at STP:

(A) 112 mL
(B) 448 mL
(C) 336 mL
(D) 450 mL

30. Down the group 13 and group 14, lower oxidation state compounds are more stable because:

(A) Down the group $I.E.$ decreases
(B) Poor shielding of electrons in $s \& p-$ orbitals as compare to d & f-orbitals
(C) Down the group electronegativity decreases
(D) Outer shell electron are less shielded from nucleus due to presence of electrons in d-orbitals of penultimate shell

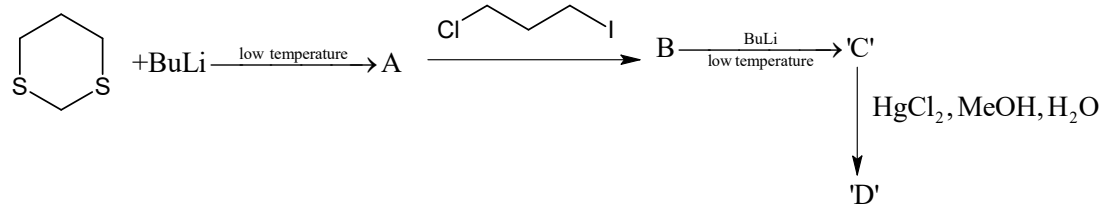
31. $2A+C \longrightarrow P$ follows the mechanism as shown below

$2A \rightleftharpoons B \quad K\left(\text{equilibrium constant}\right)$

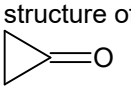
$B+C \longrightarrow P \quad K_f\left(\text{rate constant}\right)$

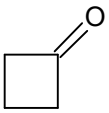
The overall rate, $+\dfrac{d[P]}{dt}$ based on above mechanism is given by

(A) $KK_f[A]^2[C][B]$
(B) $KK_f[A]^2[C]$
(C) $K[A][B]$
(D) $K_f[B][C]$

32. 

Correct structure of compound 'D' is:

(A) 
(B) CH_3CH_2CHO

(C) 

(D) $CH_3CH_2CH_2CHO$

33. Consider the Vander Waal's equation of state $\left(p+\frac{a}{V^2}\right)(V-b)=RT$ correct statements is:

(A) Value of $'a'$ of N_2 is larger than that of NH_3 due to higher molecular weight of N_2

(B) Gas 'A' has Vander Wall's constants $'a'$ $\left(L^2\,atm/mol^2\right)$ and $'b'(L/mol)$ as 7 and 4.98×10^{-2}, this gas cannot be liquid at temperature $200°C$ however high pressure may be applied.

(C) At high pressure above equation reduces to $Z=1-\frac{b}{RTV}$

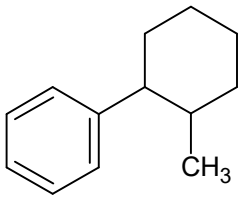
(D) If $Z<1,$ attractive forces are dominating among the gaseous molecules

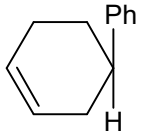
34. $\text{cis-2-phenylcyclohexanol}\xrightarrow{Na}'A'\xrightarrow{CS_2}'B'\xrightarrow{CH_3I}'C'$

$'C'$ is 2-methyl xanthate, which on pyrolysis gives:

(A) 3-phenylcyclohexene

(B) 1-phenylcyclohexene

(C) 

(D) 

35. Which of the following statement is false, if $'a'$ is the unit cell edge length?

(A) $0.866a$ is the shortest distance between cation and anion in $CsCl$ crystal.

(B) $0.707a$ is the shortest distance between two cations in CaF_2 crystal

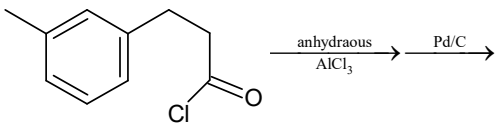
(C) $0.433a$ is the shortest distance between carbon atom in diamond

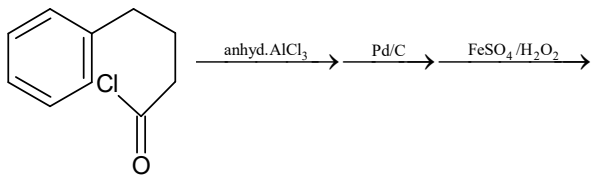
(D) $0.866a$ is the shortest distance between two cation in rock salt crystal

36. Which of the following process would result into formation of 7-methyl-1-naphthol?

(A) cis-4-(p-Tolyl)-3-butenoic acid+HF

(B) trans-4-(p-Tolyl)-3-butenoic acid+HF

(C) 

(D) 

37. When a hydrocarbon C_xH_y is burnt, gives $100ml$ of gaseous mixture of CO_2 and H_2O (vapour). On passing product mixture through anhydrous $CuSO_4$ result into decrease of volume by $60\,ml$. Which of the following lowest molecular weight hydrocarbon can satisfy the above condition?

(A) CH_4

(B) C_2H_6

(C) C_2H_4

(D) C_4H_{10}

38. While testing BO_3^{3-}, there is green-edges flame on heating the salt with conc. H_2SO_4 and CH_3OH. Green colour is of

(A) $(CH_3)_3B$

(B) $(CH_3O)_3B$

(C) B_2O_3

(D) H_3Bo_3

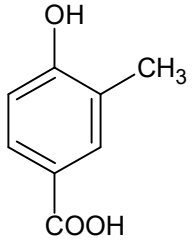
39. Vanillin $(L), C_8H_8O_3$, is isolated from vanilla beans. It give an intense colour with $FeCl_3$ and positive Tollen's test. It is not steam-distilled and does not react with HCl. It goes through following steps:

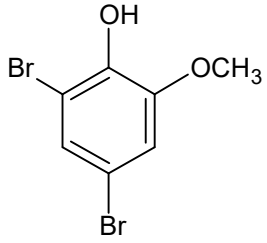
$$L \xrightarrow[1.equivalent\ of\ CH_3I]{OH^-} M,(C_9H_{10}O_3) \xrightarrow{KMnO_4} N,(C_9H_{10}O_4),(HCO_3^- \text{ soluble})$$

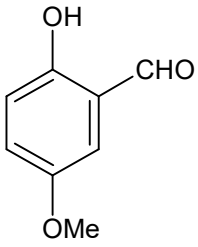
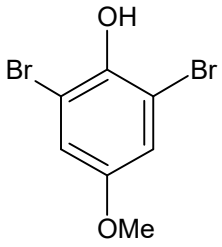
$N \xrightarrow{\text{conc. HI}\,\Delta}$ 3,4-dihydroxy benzoic acid

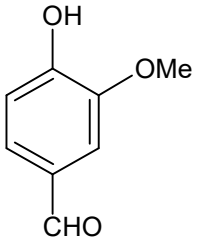
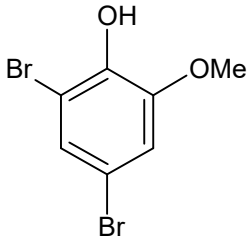
$$L \xrightarrow{[Ag(NH_3)_2]} O \xrightarrow{2\,Br_2} P,(C_7H_6Br_2O_2)$$

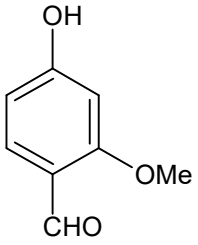
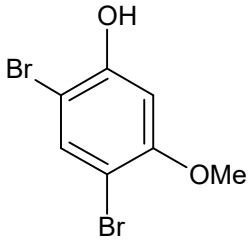
Which of the following is correct statement about various products?

(A) L is 

and P is 

(B) L is  and P is 

(C) L is  and P is 

(D) L is  and P is 

40. [X] + dil $H_2SO_4 \longrightarrow$ [Y], a colourless gas with irritating smell

[Y] + $K_2Cr_2O_7$ + $H_2SO_4 \longrightarrow$ Green solution

The species [X] and [Y] are

(A) SO_3^{2-}, SO_2

(B) Cl^-, HCl

(C) S^{2-}, H_2S

(D) CO_3^{2-}, CO_2

41. An orange coloured compound (A) when treated with $NaCl$ & conc. H_2SO_4 gives red brown vapour (B). This vapour when passed through $NaOH$ gives yellow coloured solution (C). On addition of CH_3COOH and $(CH_3COO)_2Pb$ to this yellow solution, yellow colored ppt (D) is obtained. Which of the following compound is not resemble with above information:

(A) 'A' is $K_2Cr_2O_7$

(B) 'B' is CrO_2Cl_2

(C) 'C' is Na_2CrO_4

(D) 'D' is $PbCl_2$

42. The colour of potassium dichromate is due to:

(A) d-d transition

(B) transition in K^+ ion

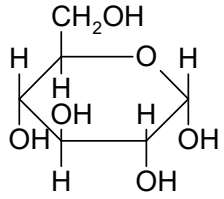
(C) ligand-to-metal charge transfer

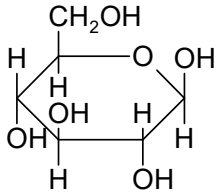
(D) metal to ligand charge transfer

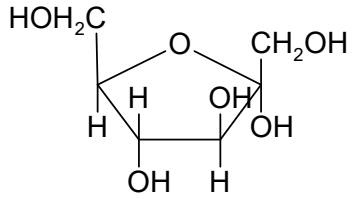
43. Adsorption of gases on solid surface is generally exothermic because:
(A) Chemical bonds are always formed
(B) Adsorbate decreases the surface energy of adsorbent
(C) Entropy increases
(D) Free energy increases

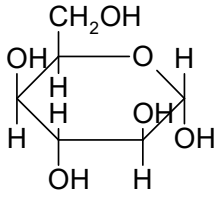
44. Incorrect statement for metal carbonyl $\left[M(CO)_6 \right]$ is:
(A) σ bonding electrons are donated by CO group to the metal.
(B) $d\pi - p\pi$ back bonding occurs from the filled metal orbitals.
(C) $d\pi - p\pi$ back bonding makes M-CO bond stronger.
(D) C – O bond length is greater in CO gas as compare to C – O bond length in given complex.

45. Which of the following is correct Haworth projection for α-D glucopyranose?

(A) 

(B) 

(C) 

(D) 

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

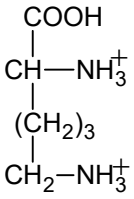
46. One electrolyzing the solution of $CH_3COONa(aq)$, the volume ratio of gases formed at anode and cathode is:

47. Distance between nearest octahedral and tetrahedral void in FCC unit cell of edge length $8\sqrt{3}$ Å is x Å then x is ____.

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48.



Lysine has $p^{ka_1}, p^{ka_2}, p^{ka_3}$ values 2.18, 8.95 and 10.53 respectively. Then value of its isoelectric point $\left(p^{I}\right)$ is:

49. A current of $0.55\,\text{Amp}$ deposits $0.55\,\text{gm}$ of a metal in 100 minutes. Calculate atomic mass of the metal if equivalent weight $= \frac{\text{M.W.}}{3}$?

50. pH of $0.01\text{M} - \left(NH_4\right)_2 SO_4$ and $0.02\text{M} - NH_4OH$ buffer $\left(p^{Ka} \text{ of } NH_4^+ = 9.26\right)$ is:

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. Standard deviation of two distributions are same and their arithmetic means are 30 and 25 respectively. If their coefficient of variation is v_1 and v_2, then $6v_1 - 5v_2$ is equal to:

(A) 0
(B) 1
(C) 2
(D) 3

52. Let $\frac{1}{3}+\frac{2}{2.3.4}+\frac{2}{3.4.5}+....+\frac{2}{n(n+1)(n+2)}=\frac{n(n+3)}{k(n+1)(n+2)}\forall nfN,$ then the value of k is:

(A) 1
(B) 2
(C) 3
(D) 4

53. A student preparing for IIT-JEE at FIITJEE is given 6 targets numbered to 1 to 6 of each subject physics, chemistry and mathematics. He or she keeps all his books in different shelves (for each subject) in order, number of ways to select all 18 books one by one such that books of each subject must be selected in order (i.e. all books of one subject to the selected one after another, then of second subject and then of third subject) is:

(A) $\frac{18!}{6!6!6!}$
(B) $\frac{18!}{6!6!6!}\times 3!$
(C) $\frac{6!6!6!}{3!}$
(D) $\frac{18!}{6!6!6!3!}$

54. If x is such that $[2\sin x]+[\cos x]=-3$, for some $x\in[0,25]$ then range of the function $g(x)=\sin x+\cos x,$ where $[.]$ represents greatest integer function is:

(A) $\left[-\sqrt{2},-1\right]$
(B) $\left[-\sqrt{2},-1\right)$
(C) $\left(-\sqrt{2},\frac{-1-\sqrt{3}}{2}\right)$
(D) $\left[\frac{-1-\sqrt{3}}{2},-1\right]$

55. If $f(x)=\{x\}+\{-x\}$ and $g(x)=\sin^{-1}x$, then number of elements in the range of $gof(x)$ is (where $\{.\}$ represents fractional part function):

(A) 0
(B) 1
(C) 2
(D) 3

56. The number of real solution of (x,y) where $|y|=\cos x$ and $y=\cos^{-1}(\cos x), -2\pi \le x \le 2\pi$ is/are:

(A) 1
(B) 2
(C) 3
(D) 4

57. Let $f(x)=[x+2]$ and $g(x)=\cos x,$ where $[.]$ denotes the greatest integer function. The value of $(fog)'\left(\frac{\pi}{2}\right)$, is:

(A) 0
(B) 1
(C) -1
(D) Does not exist

58. Number of points of non-differentiability of $f(x)=\min\{0,\cos x,\sin x\}$ in $(0,2n\pi)$ is:

(A) $3n-1$
(B) $3n$
(C) $3n+1$
(D) $3n+2$

59. If $y=\cos x$ then $\frac{d^2}{dy^2}(\sin^5 x)$ is:

(A) $15\sin x+20\sin^3 x$
(B) $15\sin x-20\sin^3 x$
(C) $20\sin x+15\sin^3 x$
(D) $20\sin x-15\sin^3 x$

60. Find the maximum possible area of rectangle such that it two vertices are on y-axis and other two vertices on the lines $y=x$ and $3x+2y=6$ respectively and the rectangle is inscribed in triangle, $y\ge 0,\ y\ge x,\ 3x+2y\le 6$ is:

(A) $\frac{3}{10}$
(B) $\frac{7}{10}$

(C) $\frac{9}{10}$

(D) $\frac{11}{10}$

61. If the line $\ell x + my + n = 0$ is a normal to the curve $xy = 1,$ then possible value of ℓ and m is:

(A) $\ell = 2, m = 1$

(B) $\ell = 2, m = -1$

(C) $\ell = -2, m = -1$

(D) $\ell = -1, m = -2$

62. In a ΔPQR (with usual solutions) if $\begin{vmatrix} 1 & p & q \\ 1 & q & r \\ 1 & r & p \end{vmatrix} = 0,$ then the value of $\cos 2P + \cos 2Q + \cos 2R$ is:

(A) $\frac{1}{2}$

(B) $-\frac{1}{2}$

(C) $\frac{3}{2}$

(D) $-\frac{3}{2}$

63. The number of values of $'t'$ for which the system of equations $(t+1)x + 8y = 4t,\ tx + (t+3)y = 3t - 1$ has no solution is:

(A) 0

(B) 1

(C) 2

(D) 3

64. $\int \frac{3x^2 + 2x}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5} dx$ is equal to:

(A) $\frac{1}{2}\tan^{-1}\left(\frac{x^3 + x^2 + 1}{2}\right) + C$

(B) $\frac{1}{2}\sin^{-1}\left(\frac{x^3 + x^2 + 1}{2}\right) + C$

(C) $\tan^{-1}\left(\frac{x^3 + x^2 + 1}{2}\right) + C$

(D) $\sin^{-1}\left(\frac{x^3 + x^2 + 1}{2}\right) + C$

65. If $\mathrm{f}\left(\mathrm{x}\right)=\int\limits_{\frac{1}{x}}^{\sqrt{x}}\sin\left(t^2\right)dt,$ then $f'\left(1\right)$ is equal to:

(A) $\frac{3}{2}\sin 1$

(B) $\frac{1}{2}\sin 1$

(C) $\sin 1$

(D) $2\sin 1$

66. Area bounded by the curves $y=\left[\frac{x^2}{36}+1\right], y=x-2$ and $x=0$ above x-axis is, where $[.]$ denotes greatest integer function:

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) $\frac{5}{2}$

(D) $\frac{7}{2}$

67. If the equation represented by the solution of the differential equation $\left(xy-1\right)\frac{dy}{dx}+y^2=0$ passes through the points $\left(2,1\right)$ and $\left(k,e\right)$, then the value of k is:

(A) e^2

(B) $\frac{1}{e}$

(C) $\frac{2}{e}$

(D) $\frac{3}{e}$

68. If a and b are selected from the set $\left\{1,2,3,.....10\right\},$ then probability that a^2-b^2 is divisible by 3 is:

(A) $\frac{1}{15}$

(B) $\frac{4}{15}$

(C) $\frac{8}{15}$

(D) $\frac{13}{15}$

69. The value of sum of the series ${}^{13}C_0.{}^{14}C_1 + {}^{13}C_1.{}^{14}C_2 + {}^{13}C_2.{}^{14}C_3 + ... + {}^{13}C_{13}.{}^{14}C_{14}$ is:

(A) ${}^{27}C_{12}$

(B) ${}^{27}C_{14}$

(C) ${}^{27}C_{16}$

(D) ${}^{27}C_{10}$

70. If A, G and H are arithmetic, geometric and harmonic means between three given numbers a, b and c then the equation having a, b and c as its roots is:

(A) $x^3 + 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0$

(B) $x^3 - 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0$

(C) $x^3 - 3Ax^2 + \frac{3G^3}{H}x + G^3 = 0$

(D) $x^3 + 3Ax^2 - \frac{3G^3}{H}x + G^3 = 0$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. Total number of unit vectors in $xy-$ plane making on angle of $45°$ with the vector $\hat{i} + \hat{j}$ and an angle of $60°$ with the vector $3\hat{i} - \hat{j}$ is:

72. If $4x - ay + 3z = 0,\ x + 2y + az = 0$ and $ax + 2z = 0$ have a non-trivial solution, then the value of a is:

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. 6 person A, B, C, D, E and F are to be seated at a circular table. In how many ways can this be done if A must always has either B or C on his right and B must always has either C or D on his right:

74. If $\mathrm{n}(\mathrm{s})$ represents the number of elements in the set S, such that $n(A) = 10$ and $n(B) = 20$, then the value of $n(A \Delta B)$ cannot be:

75. If the two circles $x^2 + y^2 = 1$ and $x^2 + y^2 - 8x - 6y + \lambda^2 = 0$ have exactly two common tangents then the number of integral value of λ is: